ATM FRAUD DETECTION SYSTEM

MAJOR PROJECT FINAL REPORT

Submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

ELECTRONICS AND COMMUNICATION ENGINEERING

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2025-2026

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CERTIFICATE

This is to certify that the major project entitled **"ATM FRAUD DETECTION SYSTEM"** Submitted by P.SIVASAIKRISHNA(23335A0414),T.VARSHITHA(22331A04HO),V.BHARGAV (22331A04H6),S.DILEEP(22331A04F3) in Partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Communication Engineering is a Bonafide work done by them under my supervision during the academic year 2025-2026.

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SELF DECLARATION

We hereby declare that the project entitled "**ATM FRAUD DETECTION SYSTEM**" is an outcome of our own efforts under the guidance of **Dr. D. Rama Devi**. The project is submitted to the Maharaj Vijayaram Gajapathi Raj College of Engineering (A), For the partial fulfilment of the Bachelor of Technology 2025-2026.We also declare that this project report has not been previously submitted to any other university.

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With great solemnity and sincerity, we offer our profuse thanks to our management, **MANSAS**, for providing all the resources to complete our project successfully.

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MISSION AND VISION

Vision of the Institution

MVGR College of Engineering strives to become a center par excellence for technical education where aspiring students can be transformed into skilled & well-rounded professionals with strong understanding of fundamentals, a flair for responsible innovation in engineering practical solutions applying the fundamentals, & confidence & poise to meet the challenges in their chosen professional spheres.

Mission of the Institution

The management believes imparting quality education in an atmosphere that motivates learning as a social obligation which we owe to the students, their parents/guardians and society at large and hence the effort is to leave no stone unturned in providing the same with all sincerity. Towards that end, the management believes special focus has to be on the following areas and provide them:

- Staff with high quality experience and facilitate lifelong learning opportunities in the field of industry and research advancements.
- A well stream-lined teaching learning process that is continuously assessed for effectiveness and fine-tuned for improvement.
- State-of-the-art lab and general infrastructure that gives students the necessary tools and means to enhance their knowledge and understanding.
- Centralized department focused on improving placement opportunities for our students directly on campus and coordinating the training programs for students to complement the curriculum and enhance their career opportunities.
- Advanced research facilities and more importantly atmosphere to encourage students to pursue self-learning on advanced topics and conduct research.

Vision of the Department

To be a center of excellence in Electronics and Communication Engineering by making students abreast with contemporary technologies meeting the standards of global industry and to develop them into skillful engineers who are technologically superior with knowledge of core as well as inter disciplinary domains with human values.

Mission of the Department

- Establish a unique learning environment to enable the students to face global challenges by using latest technologies in Electronics and Communication.
- Establish centers of excellence in niche technologies to inculcate the spirit of innovation and creativity in faculty and students.
- Provide an environment for collaborative and inter disciplinary activities that enables students to develop industry ready and entrepreneurial skills.
- Provide ethical and value-based education to serve humanity by developing socially sensitive engineers.

Program Educational Objectives (PEOs)

PEO1: Graduates will have the ability to formulate and solve problems in electronics and communication engineering domain with strong foundation in mathematics, science and engineering fundamentals.

PEO2: Graduates will have required knowledge for a successful career in the field of electronics and communication engineering.

PEO3: Graduates will have the ability to pursue higher studies and / or productively engage in research and to install a passion and systematic approach for continuous learning so that they can adapt to the evolving knowledge in their chosen profession.

PEO4: Graduates will apply professional and communication skills to function as leaders and members of multidisciplinary teams with strong organizational skills and ethical responsibility.

Program outcomes (POs)

1. An ability to apply knowledge of mathematics, science and engineering
2. An ability to design and conduct experiments, as well as to analyse and interpret data
3. An ability to design a system, component or process to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability and sustainability
4. An ability to function on multidisciplinary teams
5. An ability to identify, formulates, and solves engineering problems
6. An understanding of professional and ethical responsibility
7. An ability to communicate effectively
8. The broad education necessary to understand the impact of engineering solutions in a global, economic, environmental and societal context
9. A recognition of the need for and ability to engage in life-long learning
10. A knowledge of contemporary issues
11. An ability to use the techniques, skills and modern engineering tools necessary for engineering practice
12. An ability to design, simulate and test systems in a way that demonstrates comprehensive trade-offs involved in design choices in RF & Microwave engineering, VLSI & Embedded systems and Signal/Image processing related to industry and research activities.

Graduate Attributes:

1. Engineering Knowledge.
2. Problem Analysis.
3. Design and development of solutions.
4. Conduct investigations of complex problems.
5. Modern tool usage.
6. The Engineer and society.
7. Environment and sustainability.
8. Ethics.
9. Individual and teamwork.
10. Communication.
11. Project management and finance.
12. Life – long learning.

ABSTRACT

The increasing number of ATM fraud incidents has created a need for more secure and intelligent authentication systems. This project presents an advanced ATM fraud detection system that integrates embedded systems with machine learning techniques to enhance security. The system uses an Arduino-based hardware setup combined with RFID card verification, keypad-based PIN authentication, and real-time face recognition for multi-level security. Initially, the user scans an RFID card, followed by entering a secure PIN through a keypad. Upon successful verification, a face recognition module implemented using Python and OpenCV is activated to confirm the user's identity.

If the face matches the authorized user, access is granted; otherwise, the system identifies it as a fraudulent attempt. In such cases, a buzzer alert is triggered, an LED indication is activated, and an SMS notification is sent to the registered mobile number using a GSM module. This layered authentication approach significantly reduces the chances of unauthorized access. The proposed system is cost-effective, reliable, and suitable for real-world ATM security applications, providing an efficient solution to prevent financial fraud and enhance user Safety.

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CHAPTER-1

INTRODUCTION

Introduction to ATM Fraud Detection System:

The advancement of banking technologies has significantly improved the ease and speed of financial transactions, with Automated Teller Machines (ATMs) playing a crucial role in providing 24/7 banking services. Despite these advantages, ATM systems are increasingly becoming targets for various types of fraudulent activities such as card skimming, PIN theft, shoulder surfing, and unauthorized access. These security threats not only lead to financial losses but also reduce customer trust in banking systems.

Traditional ATM security primarily relies on card-based and PIN-based authentication methods, which are vulnerable to attacks if sensitive information is compromised. With the growing sophistication of fraud techniques, there is a need for more secure and intelligent systems that can detect and prevent fraudulent transactions effectively.

This project proposes an advanced ATM fraud detection system that integrates embedded systems with machine learning techniques to enhance security. The system utilizes components such as RFID modules, keypads, LCD displays, and possibly biometric or face recognition systems to ensure multi-level authentication. Additionally, machine learning algorithms are employed to analyze user behavior and identify unusual patterns that may indicate fraudulent activity.

The main objective of this system is to provide a secure, reliable, and efficient solution for ATM transactions by minimizing fraud risks and ensuring user authentication through multiple verification layers. By incorporating real-time monitoring and intelligent decision-making, the system aims to improve the overall security of ATM operations and protect users from potential financial threats.

CHAPTER-2

LITERATURE SURVEY

2.1 Face Recognition with PIN Authentication

A. Singh et al. (2019) proposed a face recognition-based ATM authentication system that enhances transaction security by integrating biometric verification with traditional PIN authentication. The method uses Haar Cascade for face detection, where the user's face is captured during the transaction and compared with a stored database. Access is granted only if both the PIN and facial features match, thereby reducing risks such as card theft and unauthorized usage. However, the system suffers from limitations like poor accuracy in low-light conditions and vulnerability to spoofing using photographs. To overcome these issues, the proposed work focuses on improving accuracy and robustness by using advanced techniques such as CNN/HOG-based face encoding, which perform better under varying lighting conditions and reduce spoofing risks.

2.2 IoT-Based Biometric ATM Security

R. Sharma et al. (2020) introduced a smart ATM security system that combines IoT technology with biometric authentication methods such as fingerprint recognition. The system ensures multi-level security by verifying biometric data along with PIN entry and uses GSM-based IoT modules to send real-time alerts in case of suspicious activities. While this approach enhances monitoring and communication, it has limitations including the high cost of fingerprint hardware and delays in GSM communication, especially in rural areas. To address these challenges, the proposed system aims to develop a cost-effective solution by replacing expensive fingerprint hardware with webcam-based face recognition and implementing offline alert mechanisms to reduce dependency on network connectivity.

2.3 Deep Learning-Based Face Recognition

P. Patel et al. (2021) developed a deep learning-based real-time face recognition system for ATM security using convolutional neural networks (CNNs) implemented with TensorFlow/Keras on GPU hardware. The system provides high accuracy in identifying users under different lighting conditions and facial variations, enabling secure and efficient authentication during transactions. Despite its advantages, the requirement of GPU-based processing makes the system expensive and less portable for widespread

deployment. To solve this problem, the proposed work introduces a lightweight face recognition model using HOG/CNN techniques that can run efficiently on standard systems without requiring high-end GPU resources.

2.4 IoT-Based Fraud Detection System

K. Verma et al. (2022) proposed an IoT-based fraud detection system for ATM machines that focuses on monitoring and identifying suspicious activities through various sensors and communication modules. The system continuously analyzes transaction patterns and environmental data, triggering alerts to authorities in case of abnormal behavior or intrusion attempts. Although effective in detecting physical tampering and fraud, the system lacks a mechanism for authenticating the identity of the user. To bridge this gap, the proposed system integrates multi-factor authentication methods such as RFID, PIN, and face recognition along with fraud detection, ensuring both secure user verification and enhanced protection against ATM fraud.

CHAPTER 3

PROBLEM STATEMENT

The widespread adoption of Automated Teller Machines (ATMs) has significantly improved the convenience and accessibility of banking services, enabling users to perform financial transactions at any time. However, this increased usage has also led to a rise in ATM-related fraudulent activities. Common types of fraud include card skimming, PIN theft, shoulder surfing, card cloning, and unauthorized access to user accounts. These threats pose serious challenges to the security of financial systems and can result in substantial financial losses

for both customers and banks.

Traditional ATM systems mainly depend on card-based and PIN-based authentication, which are no longer sufficient to ensure security in the face of advanced fraud techniques. Once a fraudster gains access to a user's card details and PIN, it becomes relatively easy to carry out unauthorized transactions. Additionally, existing systems lack the capability to analyze transaction patterns or detect abnormal user behavior in real time. This limitation reduces the effectiveness of fraud prevention mechanisms and increases the risk of undetected fraudulent activities.

Furthermore, many ATM systems do not incorporate intelligent decision-making or multi-layered authentication processes, making them more vulnerable to attacks. There is a growing need for a system that not only authenticates users securely but also continuously monitors transactions to identify suspicious activities.

This project addresses these challenges by proposing an ATM fraud detection system that integrates machine learning with embedded systems. The system is designed to provide enhanced security through multiple verification layers and real-time analysis of user behavior. By detecting anomalies and preventing unauthorized access, the proposed solution aims to reduce fraud, protect user data, and improve the overall reliability and trustworthiness of ATM transactions.

CHAPTER 4

OBJECTIVES OF THE PROJECT

4.1 To Enhance ATM Security

The primary objective of this project is to improve the overall security of ATM transactions. Traditional systems rely only on ATM cards and PINs, which can be easily compromised. This project introduces additional security layers to make unauthorized access more difficult.

4.2 To Implement Multi-Level Authentication

This system aims to provide multiple levels of user verification such as RFID/card verification, PIN entry, and possibly biometric or face recognition. Multi-level authentication ensures that even if one security layer is breached, the system can still prevent fraud.

4.3 To Detect Fraud Using Machine Learning

Another key objective is to use machine learning algorithms to analyze user behavior and transaction patterns. The system can identify unusual or suspicious activities (such as abnormal withdrawal patterns) and flag them as potential fraud in real time.

4.4 To Integrate Embedded Systems with ML

The project focuses on combining hardware components (like Arduino, RFID, keypad, LCD) with software (Python and ML models). This integration helps in creating a smart system capable of both physical authentication and intelligent decision-making.

4.5 To Provide Real-Time Monitoring and Alerts

The system is designed to monitor transactions continuously and provide instant alerts (via display or GSM module) if any suspicious activity is detected. This helps in taking immediate action to prevent fraud.

4.6 To Improve User Trust and Reliability

By enhancing security and reducing fraud risks, the system aims to increase user confidence in ATM transactions. A secure system ensures safe banking experiences for users.

4.7 To Develop a Cost-Effective Solution

The project uses affordable components like Arduino and basic sensors, making it a cost-effective solution that can be implemented without high investment

CHAPTER 5

PROPOSED SYSTEM

The proposed system aims to develop a secure and intelligent ATM fraud detection mechanism by integrating embedded systems with machine learning techniques. Unlike traditional ATM systems that rely only on card and PIN authentication, the proposed system introduces multi-level security to enhance protection against fraudulent activities.

In this system, the user initiates the transaction by scanning an RFID card, which acts as a replacement or enhancement to the conventional ATM card. Once the card is verified, the user is prompted to enter a PIN through a keypad. The entered PIN is validated within the embedded system (Arduino). If the authentication is successful, the system proceeds to the next level of verification.

To further strengthen security, the system can incorporate an additional authentication method such as face recognition or OTP verification. After successful user authentication, the transaction details and user behavior are sent to a machine learning model developed in Python. The ML model analyzes parameters such as transaction amount, frequency, and usage patterns to determine whether the transaction is legitimate or potentially fraudulent.

If the system detects any abnormal or suspicious activity, it immediately blocks the transaction and generates an alert message, which can be displayed on the LCD screen or sent via a GSM module to the user or bank authorities. If no fraud is detected, the transaction is allowed to proceed normally.

This proposed system ensures real-time fraud detection, multi-layer authentication, and improved decision-making capabilities. By combining hardware and intelligent software, it provides a robust, reliable, and efficient solution to minimize ATM fraud and enhance overall banking security.

5.1 System Components

The system consists of hardware components such as RFID module, keypad, Arduino microcontroller, LCD display, and GSM module. The software components include Arduino IDE for embedded programming and Python for implementing machine learning algorithms

5.2 Authentication Process

The process begins with RFID card verification followed by PIN entry through the keypad The system checks whether the entered PIN matches the stored data. Additional authentication

methods like OTP or face recognition can also be included.

5.3 Machine Learning-Based Fraud Detection

The system uses a machine learning model to analyze transaction details such as user behavior, transaction amount, and frequency. It helps in identifying unusual patterns and detecting fraudulent activities.

5.4 Data Flow in the System

User input is collected through hardware components and processed by Arduino. The data is sent to the machine learning model, which analyzes it and returns the result to the system for further action.

5.5 Alert and Response Mechanism

If fraud is detected, the system blocks the transaction and sends alerts through the LCD display and GSM module to inform the user or authorities.

5.6 Advantages of the Proposed System

The system provides enhanced security, real-time monitoring, multi-level authentication, and improved reliability compared to traditional ATM systems.

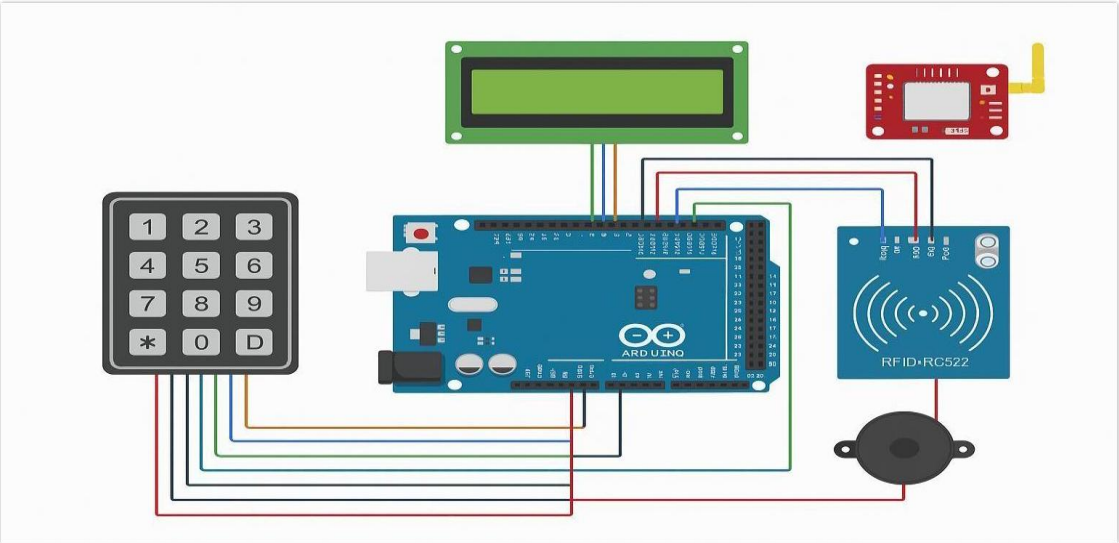

Figure 1

Fig 5.1: System Architecture

CHAPTER 6
SYSTEM ARCHITECTURE

The system architecture of the proposed ATM fraud detection system is designed to provide a secure, reliable, and intelligent framework by integrating embedded hardware components with machine learning techniques. It defines the structure and interaction between different modules involved in authentication, data processing, fraud detection, and response mechanisms. The architecture ensures smooth communication between hardware and software components, enabling real-time monitoring and accurate decision-making to prevent unauthorized transactions.

The hardware architecture consists of essential components such as an RFID module for card verification, a keypad for secure PIN entry, an Arduino microcontroller that acts as the central processing unit, an LCD display for user interaction, and a GSM module for sending alerts. These components are interconnected, with the Arduino coordinating all operations, collecting user inputs, and controlling outputs.

The software architecture includes embedded programming using Arduino IDE and a machine learning model developed in Python. Once the user successfully completes authentication, the transaction details are sent to the machine learning module. The model analyzes parameters such as transaction amount, frequency, and behavioral patterns to detect any anomalies or suspicious activities.

The communication module ensures seamless data transfer between the Arduino and the machine learning system through serial communication. Based on the analysis results, the system decides whether to approve or block the transaction. In case of fraud detection, alerts are generated immediately.

Overall, this architecture provides a multi-layered security approach, enhances system efficiency, and ensures real-time fraud detection, making ATM transactions safer and more reliable.

6.1 Hardware Architecture

The hardware architecture consists of physical components that interact directly with the user. These include the RFID module for card scanning, keypad for PIN entry, Arduino microcontroller for processing, LCD display for user interface, and GSM module for communication. Each component is connected to the Arduino, which acts as the central control unit. The hardware components work together to capture user inputs and provide outputs based on system decisions.

6.2 Software Architecture

The software architecture includes embedded programming and machine learning modules. The Arduino is programmed using Embedded C in the Arduino IDE to control hardware operations such as reading inputs and displaying outputs. On the other hand, Python is used to implement machine learning

algorithms that analyze transaction data. The software ensures proper coordination between hardware and intelligent decision-making systems.

6.3 Input Module

The input module is responsible for collecting data from the user. It includes the RFID module, which reads the user's card details, and the keypad, which allows the user to enter the PIN. These inputs are sent to the Arduino for verification. The accuracy and reliability of input data are crucial for the proper functioning of the system.

6.4 Processing Module

The processing module is handled by the Arduino microcontroller. It verifies the RFID data and PIN entered by the user. After successful authentication, it forwards the transaction details to the machine learning model. The Arduino acts as a bridge between hardware inputs and software analysis.

6.6 Communication Module

The communication module ensures data exchange between the Arduino and the machine learning system using serial communication. Additionally, the GSM module is used to send alerts or notifications to users or authorities in case of suspicious activity. This module plays a key role in real-time data transmission.

6.7 Output Module

The output module provides feedback to the user. The LCD display shows messages such as "Enter PIN," "Access Granted," or "Fraud Detected." If fraud is identified, the system blocks the transaction and sends alerts. This module ensures that the user is informed about the system status.

6.8 Security and Decision-Making Layer

This layer combines authentication results and machine learning predictions to make final decisions. Based on the analysis, the system either allows the transaction or blocks it. This ensures multi-layer security and reduces the chances of fraud.

6.9 Data Flow Architecture

The data flow starts from user input through RFID and keypad, moves to Arduino for processing, and then to the machine learning model for analysis. The result is sent back to the Arduino, which then controls the output devices. This structured flow ensures efficient and accurate system operation.

6.10 System Integration

The integration of all modules ensures seamless operation of the system. Hardware and software components are interconnected to provide a unified solution. Proper integration improves system performance, reliability, and scalability

CHAPTER 7
HARDWARE COMPONENTS

- Arduino Microcontroller
- RFID Module
- RFID Cards/Tags
- Keypad (4x4)
- LCD Display (16x2)
- GSM Module
- Buzzer
- LED
- Connecting Wires
- Breadboard

7.1 Arduino Microcontroller

The Arduino Uno is a widely used open-source microcontroller board based on the ATmega328P microcontroller. It serves as the central processing unit (CPU) of the proposed ATM fraud detection system, controlling and coordinating all the hardware and software operations. The Arduino Uno is responsible for receiving input data from various components such as the RFID module and keypad, processing this data according to the programmed logic, and generating appropriate outputs.

In this project, the Arduino Uno plays a crucial role in implementing the authentication process. When a user scans the RFID card, the Arduino reads the unique ID and verifies it against stored data. It then prompts the user to enter a PIN using the keypad. The entered PIN is compared with the stored credentials to determine whether access should be granted or denied. This ensures the first level of security in the system.

Additionally, the Arduino Uno communicates with the machine learning module developed in Python through serial communication. It sends transaction-related data such as authentication status and user inputs to the ML model for further analysis. Based on the response received, the Arduino decides whether to allow the transaction or block it.

The Arduino also controls output devices such as the LCD display, buzzer, LED indicators, and GSM module. It displays messages to guide the user, activates alerts in case of suspicious activity, and ensures real-time system response. Its ease of programming, flexibility, and compatibility with various modules make it an ideal choice for embedded system applications like ATM fraud detection

Figure 2

Fig 7.1:Arduino

7.2 RFID Module

RFID (Radio Frequency Identification) is a wireless technology used for automatic identification and data capture. It uses radio waves to communicate between an RFID reader (module) and an RFID tag or card. An RFID system mainly consists of two parts: the RFID reader and the RFID tag. The reader emits radio signals, which are received by the tag, and the tag responds by sending its stored data, usually a unique identification number (UID). RFID technology is widely used in applications such as access control systems, contactless payments, inventory tracking, and security systems due to its fast and contactless operation. It does not require direct line-of-sight like barcode systems, making it more efficient and reliable.

In the proposed ATM fraud detection system, the RFID module plays a crucial role in enhancing security. It is used as the first level of authentication by verifying the user's card. When a user scans the RFID card, the module reads the unique ID and sends it to the Arduino for verification. Only if the card is recognized as valid, the system allows the user to proceed to the next step, which is entering the PIN. This helps in preventing unauthorized users from accessing the system.

The use of RFID improves the overall security by reducing risks associated with traditional ATM cards such as card skimming and duplication. Since each RFID card has a unique identification number, it becomes difficult for fraudsters to replicate it. Additionally, the contactless feature ensures quick and smooth operation, making the system more efficient. Thus, the RFID module serves as a reliable and secure method for initiating user authentication in the ATM fraud detection system.

Figure 3

Fig 7.2: **RFID Module**

7.3 RFID Cards/Tags

RFID tags are small electronic devices used in RFID systems for identification and data storage. Each RFID tag contains a microchip and an antenna. The microchip stores a unique identification number (UID), while the antenna helps in transmitting this data to the RFID reader using radio frequency signals. RFID tags are generally of two types: passive and active. Passive RFID tags do not have their own power source and are powered by the electromagnetic field generated by the RFID reader, whereas active tags have an internal battery and can transmit signals over longer distances. In most embedded applications like this project, passive RFID tags are commonly used due to their low cost, compact size, and ease of use.

In the ATM fraud detection system, RFID tags are used as user identification cards. Each user is assigned a unique RFID tag that acts as a digital identity. When the user places the tag near the RFID module, the stored UID is transmitted to the reader. The Arduino receives this UID and checks whether it matches the registered data in the system. If the tag is recognized as valid, the system proceeds to the next level of authentication, which is PIN entry.

The use of RFID tags enhances security by ensuring that only authorized users can access the system. Since each tag has a unique code, it is difficult to duplicate or forge. Additionally, the contactless operation makes the system fast and convenient for users. RFID tags, therefore, play a crucial role in initiating secure authentication and preventing unauthorized access in the ATM fraud detection system.

Figure 4

Fig 7.3:RFID Tags

7.4 Keypad (4x4)

A 4×4 keypad is an input device commonly used in embedded systems for user interaction. It consists of 16 keys arranged in a matrix of 4 rows and 4 columns. Each key represents a number (0–9), alphabets (A–D), or special characters (* and #). The keypad works on a matrix principle, where rows and columns are connected to the microcontroller. When a key is pressed, a connection is established between a specific row and column, allowing the microcontroller to identify which key has been pressed.

The 4×4 keypad is widely used in applications such as security systems, password-based access, calculators, and embedded control systems due to its simplicity, reliability, and ease of interfacing with microcontrollers like Arduino. It requires fewer input pins because of its matrix structure, making it efficient for hardware design.

In the ATM fraud detection system, the keypad is used for entering the Personal Identification Number (PIN). After the RFID card is verified, the system prompts the user to input their PIN using the keypad. The entered PIN is then sent to the Arduino, where it is compared with the stored PIN for authentication. If the PIN matches, the system proceeds to the next stage; otherwise, access is denied.

The keypad plays a crucial role in providing the second level of security. Even if an unauthorized person gains access to the RFID card, they cannot complete the transaction without knowing the correct PIN. Additionally, the system can be programmed to limit the number of incorrect attempts and trigger alerts using a buzzer or GSM module. Thus, the 4×4 keypad enhances security, ensures user authentication, and makes the system more reliable.

Figure 5

Fig7.4:Keypad

7.5 LCD Display (16x2)

A 16×2 LCD (Liquid Crystal Display) is a commonly used output device in embedded systems for displaying text information. It consists of 16 columns and 2 rows, allowing it to display up to 32 characters at a time. The display works by controlling liquid crystals to form visible characters when electrical signals are applied. It is widely used due to its low cost, low power consumption, and easy interfacing with microcontrollers like Arduino.

The LCD can operate in 4-bit or 8-bit mode and is usually connected to a microcontroller using digital pins or through an I2C module, which reduces the number of required connections. It is capable of displaying alphabets, numbers, and some special characters, making it suitable for user interface applications.

In the ATM fraud detection system, the LCD display plays a crucial role as the main communication interface between the system and the user. It provides real-time instructions and feedback during the transaction process. For example, it displays messages such as “Scan Card,” “Enter PIN,” “Access Granted,” or “Fraud Detected.” These messages guide the user step-by-step, ensuring smooth interaction with the system.

Additionally, the LCD helps in displaying alerts and system status, especially when suspicious activity is detected. It enhances user experience by providing clear and immediate information. Without the LCD, users would not be able to understand the system’s responses easily. Therefore, the LCD display is essential for improving usability, communication, and overall functionality of the ATM fraud detection system.

Figure 6

Fig 7.5:LCD Display

7.6 GSM Module

A GSM (Global System for Mobile Communication) module is a communication device used to send and receive data over a mobile network. It operates using a SIM card, similar to a mobile phone, and enables devices to communicate through SMS, calls, or internet services. GSM modules such as SIM800 or SIM900 are commonly used in embedded systems due to their reliability and wide network coverage. They communicate with microcontrollers like Arduino using serial communication (TX and RX pins), allowing easy integration.

The GSM module is widely used in applications such as remote monitoring systems, home automation, vehicle tracking, and security systems because it enables long-distance wireless communication without the need for internet connectivity.

In the ATM fraud detection system, the GSM module plays a crucial role in providing real-time alerts and notifications. When the system detects suspicious activity, such as incorrect PIN attempts, unauthorized access, or fraudulent transactions identified by the machine learning model, the Arduino triggers the GSM module to send an alert message.

For example, the system can send an SMS to the user or bank authorities stating that a suspicious transaction has been detected or blocked. This allows immediate action to be taken, such as blocking the account or verifying the transaction. The GSM module ensures that the alert reaches the concerned person instantly, even if they are far away.

Thus, the GSM module enhances the security of the system by enabling real-time communication, quick response, and remote monitoring, making the ATM fraud detection system more reliable and effective.

Figure 7

Fig7.6 GSM Module

7.7 Buzzer

A buzzer is an electronic audio output device used to generate sound signals in embedded and electronic systems. It is commonly used to provide audible alerts, warnings, or notifications based on system conditions. Buzzers are broadly classified into two types: active buzzers and passive buzzers. An active buzzer contains an internal oscillator circuit, which means it can produce sound automatically when power is supplied. In contrast, a passive buzzer requires an external signal (usually from a microcontroller) to generate different tones or frequencies. Due to its simple design, low power consumption, compact size, and ease of interfacing, the buzzer is widely used in applications such as alarm systems, timers, security devices, and user notification systems.

In the ATM fraud detection system, the buzzer plays a crucial role in enhancing security by providing immediate and noticeable audio alerts. It is directly connected to the Arduino microcontroller, which controls its operation based on predefined conditions. For instance, when a user enters an incorrect PIN, the buzzer generates a short warning sound. If multiple incorrect attempts are made, the buzzer may produce continuous or repeated sounds to indicate a potential security threat.

Additionally, when the machine learning model detects suspicious or fraudulent activity, the Arduino activates the buzzer to alert nearby users or authorities instantly. This real-time audio indication ensures that any unusual activity does not go unnoticed, even if the user is not actively looking at the display. The buzzer can also be programmed to produce different sound patterns for different events, such as successful authentication, failed attempts, or fraud detection.

The buzzer works in coordination with other components like the LCD display and LED indicators to provide a complete alert system. While the LCD provides visual information and LEDs give quick status indication, the buzzer ensures that alerts are heard clearly. This combination significantly improves the responsiveness and effectiveness of the system.

Figure 8

Fig 7.7:Buzzer

7.8 LED

An LED (Light Emitting Diode) is a semiconductor electronic component that emits light when an electric current passes through it. It is widely used in electronic and embedded systems as an indicator due to its low power consumption, long lifespan, fast response time, and compact size. LEDs are available in different colors such as red, green, and yellow, each typically representing different statuses or conditions in a system. They are easy to interface with microcontrollers like Arduino using simple digital output pins. In the ATM fraud detection system, LEDs are used as visual indicators to display the status of the system and provide immediate feedback to the user. They are connected to the Arduino and are controlled based on different conditions during the transaction process. For example, a green LED can be used to indicate successful authentication or that the transaction is allowed, while a red LED can indicate an error, incorrect PIN entry, or detection of fraudulent activity.

When the user scans a valid RFID card and enters the correct PIN, the green LED glows, indicating that the system has granted access. On the other hand, if the user enters an incorrect PIN or if the machine learning model detects suspicious behavior, the red LED turns on, signaling that the transaction is blocked. LEDs can also be programmed to blink in specific patterns to indicate different system states, such as multiple failed attempts or system warnings.

The LED works alongside other components like the LCD display and buzzer to create a complete alert

system. While the LCD provides detailed messages and the buzzer gives audio alerts, LEDs offer quick and clear visual feedback that can be easily understood at a glance.

Thus, LEDs play an important role in enhancing user interaction, improving system clarity, and strengthening the overall security mechanism of the ATM fraud detection system by providing instant visual indications of system status.

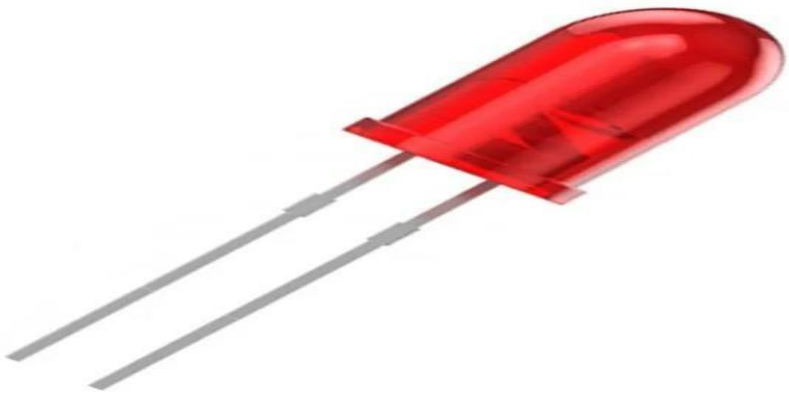

Figure 9

Fig 7.8:LED

7.9 Connecting Wires

Connecting wires are essential components in any electronic and embedded system, used to establish electrical connections between different hardware components. They are typically made of conductive materials such as copper and are insulated with plastic to prevent short circuits and ensure safe operation. Connecting wires come in different types, such as jumper wires (male-to-male, male-to-female, and female-to-female), and are widely used in prototyping and circuit design. They help in transmitting electrical signals and power between components like sensors, microcontrollers, and output devices.

In the ATM fraud detection system, connecting wires play a crucial role in linking all hardware components such as the Arduino microcontroller, RFID module, keypad, LCD display, GSM module, buzzer, and LED indicators. These wires ensure proper communication and data transfer between input devices (like RFID and keypad), processing unit (Arduino), and output devices (LCD, buzzer, LEDs).

Without proper connections, the system would not function correctly, as data and control signals would not reach the intended components.

Connecting wires also help in distributing power from the power supply to all modules, ensuring that each component receives the required voltage for operation. Proper wiring is important to avoid loose connections, signal loss, or short circuits, which can affect system performance and reliability.

During the development phase, connecting wires are especially useful because they allow easy assembly, modification, and troubleshooting of the circuit without the need for permanent soldering. This flexibility helps in testing different configurations and correcting errors efficiently.

Thus, connecting wires, though simple, are a vital part of the ATM fraud detection system, ensuring seamless connectivity, reliable data transfer, and proper functioning of all integrated components.

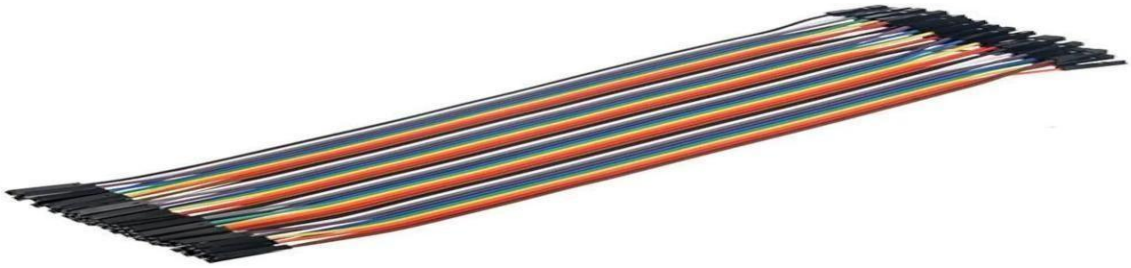

Figure 10

Fig 7.9:Connecting Wires

7.10 Breadboard

A breadboard is a reusable electronic prototyping device used to design, build, and test circuits without the need for soldering. It consists of a plastic base with a grid of small holes, internally connected by conductive metal strips. These connections allow electronic components and wires to be easily inserted and interconnected. A small breadboard is compact and suitable for simple and medium-sized projects. It also includes power rails on the sides, which are used to supply voltage (VCC) and ground (GND) to different parts of the circuit.In the ATM fraud detection system, the breadboard plays an important role in assembling and connecting various hardware components such as the RFID module, keypad, LCD display, Arduino, buzzer, and LED indicators. It provides a flexible platform where components can be connected and rearranged easily during development. This is especially useful for testing and debugging the system, as connections can be modified quickly without causing any permanent changes.The breadboard also helps in organizing the circuit neatly, reducing the chances of short circuits and wiring errors. It allows users to experiment with different configurations and identify faults efficiently. Since no soldering is required, components can be reused multiple times, making it cost-effective and convenient.

Thus, the small breadboard simplifies circuit design, enhances flexibility, and supports efficient development of the ATM fraud detection system

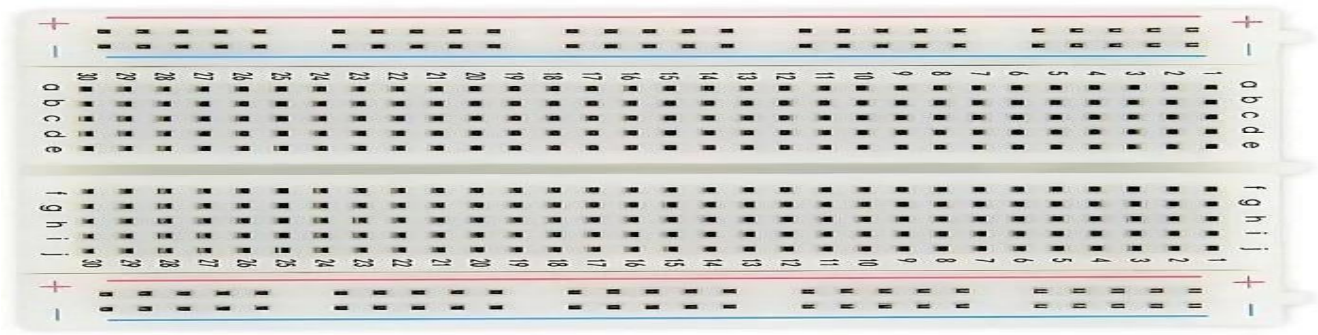

Figure 11

Fig 7.10:Breadboard

CHAPTER 8

SOFTWARE COMPONENTS

The software components of the ATM fraud detection system are responsible for controlling the hardware, processing user inputs, and implementing intelligent fraud detection. They act as the backbone of the system by enabling communication between different modules and ensuring smooth operation. These components include programming environments, coding languages, and machine learning algorithms that work together to perform authentication, analyze transaction data, and make decisions in real time. By integrating embedded programming with machine learning, the software components enhance the system's ability to detect suspicious activities, improve accuracy, and ensure secure and efficient ATM transactions.

SOFTWARE COMPONENTS USED:

8.1 Embedded C

8.2 Arduino IDE

8.3 Python

8.4 Face Detection Using HOG

8.5 Face Recognition Using CNN

8.6 Face Encoding and Comparison

8.7 Serial Communication

8.8 Libraries Used

- 8.8.1 OpenCV
- 8.8.2 face_recognition
- 8.8.3 NumPy
- 8.8.4 Pickle
- 8.8.5 PySerial
- 8.8.6 MFRC522
- 8.8.7 Keypad
- 8.8.8 LiquidCrystal_I2C
- 8.8.9 SoftwareSerial

8.1 Embedded C

Embedded C is a specialized programming language used for developing software for embedded systems, particularly microcontrollers like the Arduino Uno. It is an extension of the standard C language with additional features that allow direct interaction with hardware components such as sensors, input devices,

and output modules. Embedded C provides low-level access to memory, input/output ports, and registers, making it highly efficient for real-time applications where speed and accuracy are critical.

One of the key advantages of Embedded C is its ability to provide precise control over hardware operations. It allows the programmer to write instructions that directly control devices like RFID modules, keypads, LCD displays, buzzers, and LEDs. This makes it ideal for systems where timing and synchronization are important. Additionally, Embedded C is lightweight, consumes less memory, and executes faster compared to high-level languages, which is essential for microcontroller-based systems with limited resources.

In the ATM fraud detection system, Embedded C is used to develop the core program that runs on the Arduino microcontroller. It handles all the hardware interactions and controls the overall workflow of the system. For example, it reads the unique ID from the RFID module when a card is scanned, processes the input from the keypad when the user enters a PIN, and verifies the entered credentials with stored data. Based on the verification results, it controls output devices such as displaying messages on the LCD, activating the buzzer for alerts, and turning LEDs on or off.

Furthermore, Embedded C enables communication between the Arduino and the machine learning module using serial communication. It sends transaction-related data to the Python-based ML model and receives the analysis results. Depending on whether the transaction is classified as genuine or fraudulent, the program takes appropriate action, such as allowing the transaction or blocking it and triggering alerts.

Embedded C also allows implementation of security features like limiting the number of incorrect PIN attempts, generating warnings, and activating emergency alerts. Its reliability, efficiency, and ability to directly control hardware make it a crucial component in developing a secure and responsive ATM fraud detection system.

8.2 Arduino IDE

Arduino IDE (Integrated Development Environment) is a software platform used to write, edit, compile, and upload programs to Arduino boards such as Arduino Uno. It provides a simple and user-friendly interface, making it suitable for both beginners and advanced users in embedded system development. The IDE supports programming in Embedded C/C++ and includes built-in libraries that simplify interfacing with hardware components like sensors, displays, and communication modules.

The Arduino IDE consists of a code editor, compiler, and uploader. The code written in the IDE is called a “sketch.” Once the program is written, the IDE compiles it to check for errors and then uploads it to the Arduino microcontroller via a USB connection. It also provides a Serial Monitor feature, which allows real-time communication between the computer and Arduino for debugging and testing purposes.

In the ATM fraud detection system, Arduino IDE is used to develop and manage the embedded program that controls all hardware components. The programmer writes code in the IDE to handle operations such as reading data from the RFID module, detecting key presses from the keypad, verifying the user's PIN, and displaying messages on the LCD. It also includes logic for controlling the buzzer and LED indicators based on system conditions.

Additionally, Arduino IDE is used to implement serial communication between the Arduino and the machine learning model developed in Python. It helps in sending user authentication data to the ML system and receiving the results for further action. The Serial Monitor is especially useful for testing and debugging the system by displaying real-time data and system responses.

Thus, Arduino IDE plays a vital role in programming, testing, and controlling the embedded part of the ATM fraud detection system, ensuring accurate operation and smooth integration of all components.

Figure 12

Fig 8.1:Arduino IDE

8.3 Python

Python is a high-level, versatile programming language widely used for software development, data analysis, and machine learning applications. It is known for its simple syntax, readability, and extensive library support, which makes it easy to develop complex systems efficiently. Python provides powerful libraries such as NumPy, Pandas, Scikit-learn, and TensorFlow that are commonly used for data processing and implementing machine learning algorithms.

In the ATM fraud detection system, Python is mainly used to implement the machine learning module. It is responsible for analyzing transaction data and identifying whether a transaction is normal or fraudulent. The system collects data such as user behavior, transaction amount, frequency, and patterns, which are then processed using Python. The machine learning model is trained using this data to recognize patterns of genuine and suspicious activities.

Once the model is trained, Python is used to make real-time predictions. When the Arduino sends

transaction-related data through serial communication, the Python program receives it and passes it to the machine learning model. The model then analyzes the input and classifies the transaction as either legitimate or fraudulent. Based on this result, Python sends a response back to the Arduino.

Additionally, Python helps in preprocessing data, removing noise, and improving the accuracy of predictions. It also allows easy modification and upgrading of the system by changing or improving the machine learning model.

Thus, Python plays a critical role in adding intelligence to the ATM fraud detection system by enabling real-time analysis, accurate fraud detection, and efficient decision-making, making the system more secure

8.4 Face Detection Using HOG

Histogram of Oriented Gradients (HOG) is a feature extraction technique used in computer vision for detecting objects, especially human faces. It works by analyzing the distribution of intensity gradients or edge directions in an image. The image is divided into small regions called cells, and for each cell, the gradient directions are calculated and stored in the form of a histogram. These histograms represent the shape and structure of objects in the image. By combining these features, HOG can effectively identify the presence of a face in an image. It is widely used because it is simple, fast, and does not require heavy computational resources compared to deep learning methods.

In the ATM fraud detection system, HOG is used for face detection as the first step in the image processing pipeline. When the camera captures a frame, the system uses HOG-based detection to locate the face within the image. This is done using functions like face_recognition.face_locations(), which internally uses HOG to identify the position of faces in the frame.

Once the face is detected, the system extracts the relevant portion of the image and passes it to the next stage, which is face recognition using a CNN-based model. Without proper face detection, the system would not be able to accurately recognize the user. Thus, HOG plays a crucial role in identifying and isolating the face from the background.

By efficiently detecting faces in real time, HOG improves the speed and reliability of the system, ensuring that only valid facial data is processed for authentication. This enhances the overall accuracy and performance of the ATM fraud detection system.

8.5 Face Recognition Using CNN

Convolutional Neural Network (CNN) is a deep learning algorithm specifically designed for processing images and recognizing patterns. It consists of multiple layers such as convolutional layers, pooling layers, and fully connected layers. These layers work together to automatically extract important features like edges, textures, and facial structures from images. Unlike traditional methods, CNN does not require

manual feature extraction, as it learns features directly from the data during training. This makes it highly accurate and reliable for tasks such as face recognition.

In general, CNN-based face recognition systems convert a face image into a numerical representation known as a feature vector or encoding. This encoding captures unique facial characteristics such as the distance between eyes, shape of the nose, and overall face structure. These encodings are then compared with stored encodings to identify whether the face matches a known person.

In the ATM fraud detection system, CNN is used for recognizing the user's face after it is detected using HOG. When the camera captures the user's face, the system processes the image and generates a face encoding using a CNN-based model (through the face_recognition library). These encodings are compared with the pre-stored encodings saved in a file (encodings.pkl).

If the similarity between the captured face and stored data is within a defined threshold, the system identifies the user as authorized and sends a "GRANT" signal to the Arduino. Otherwise, it sends a "DENY" signal, indicating a possible unauthorized access attempt.

Thus, CNN plays a crucial role in providing an additional layer of security by accurately verifying the identity of the user, making the ATM fraud detection system more robust, intelligent, and secure.

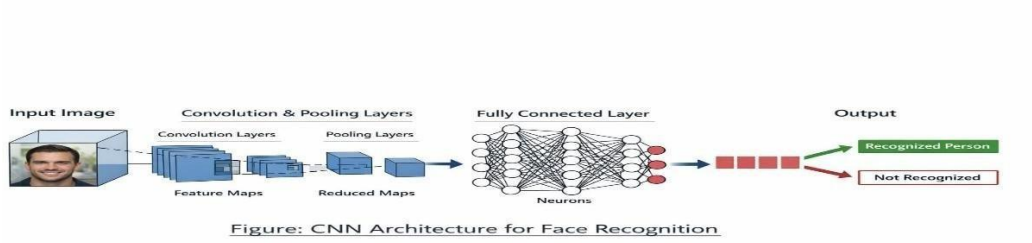

Figure 13

Fig 8.2:CNN Architecture For Face Recognition

8.6 Face Encoding and Comparison

Face encoding is the process of converting a human face into a numerical representation that captures its unique features. Instead of working directly with images, the system transforms each detected face into a vector of numbers (typically a 128-dimensional feature vector). This encoding contains important facial characteristics such as the distance between eyes, shape of the nose, jawline structure, and other distinguishing features. These numerical representations make it easier and faster to compare faces accurately.

Face comparison is the next step, where the encoded face of the current user is compared with previously stored encodings in the system. This comparison is usually done using a distance metric such as Euclidean

distance. If the distance between the two encodings is small, it means the faces are similar (same person). If the distance is large, the faces are considered different. A predefined threshold value is used to decide whether the match is acceptable or not.

In the ATM fraud detection system, face encoding and comparison play a crucial role in verifying the identity of the user. After the face is detected using HOG and processed using a CNN-based model, the system generates an encoding for the captured face. This encoding is then compared with the stored encodings loaded from a file (encodings.pkl), which contains data of authorized users.

If the distance between the captured encoding and any stored encoding is less than the defined threshold (for example, 0.5), the system considers it a match and sends a "GRANT" signal to the Arduino. Otherwise, it sends a "DENY" signal, indicating a possible unauthorized access attempt.

Thus, face encoding and comparison enable accurate and efficient identity verification, forming a critical part of the ATM fraud detection system by ensuring that only authorized users are allowed access.

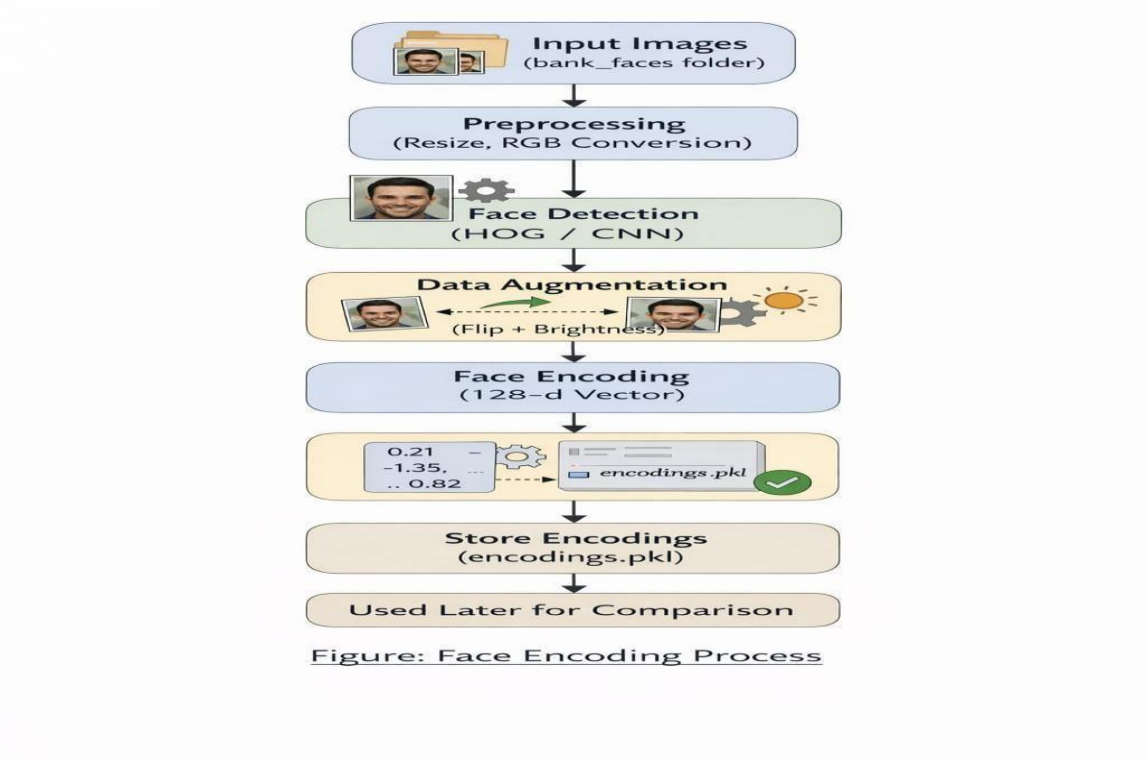

Figure 14 ***Fig 8.3 Face Encoding Process***

8.7 Serial Communication

Serial communication is a method of transmitting data between devices one bit at a time over a communication channel. In embedded systems, it is commonly used to enable communication between a microcontroller and a computer. It operates using protocols such as UART (Universal Asynchronous Receiver Transmitter), where data is sent through two main lines: Transmit (TX) and Receive (RX). In

Arduino-based systems, serial communication is implemented using functions like Serial.begin(), Serial.print(), and Serial.read(), while in Python it is handled using the PySerial library.

In the ATM fraud detection system, serial communication plays a crucial role in connecting the Arduino microcontroller with the Python-based face recognition system. The Arduino is connected to the computer using a USB cable, which provides both power and a communication interface. Over this connection, serial communication is established to exchange data between the two systems.

The process works as follows: after successful RFID card verification and PIN entry, the Arduino sends a command such as "START_FACE" to the Python program through serial communication. The Python program receives this command, activates the camera, and performs face detection and recognition using HOG and CNN. Once the verification process is complete, Python sends back the result as either "GRANT" or "DENY".

The Arduino then reads this response and takes appropriate action. If access is granted, it allows the transaction and displays a success message. If access is denied, it activates the buzzer and sends an alert message using the GSM module.Thus, serial communication acts as a bridge between hardware and software, enabling real-time data exchange and ensuring smooth coordination between the embedded system and the machine learning module.

8.8 Libraries Used

Libraries are pre-written collections of code that help in performing specific tasks easily without writing everything from scratch. In this project, various libraries are used to handle image processing, face recognition, hardware communication, and data storage efficiently.

8.8.1 OpenCV (cv2)

Used for image and video processing. It captures real-time video from the camera, converts image formats, and displays output such as "GRANTED" or "DENIED".

8.8.2 face_recognition

Used for face detection and recognition. It detects faces, generates face encodings, and compares them with stored data to verify identity.

8.8.3 NumPy

Used for numerical operations. It helps in handling arrays and performing calculations required for face encoding and comparison.

8.8.4 Pickle

Used for data storage. It stores and loads face encodings in a file (encodings.pkl) for future use.

8.8.5 PySerial

Used for serial communication between Arduino and Python. It sends and receives data like "START_FACE", "GRANT", and "DENY".

8.8.6 MFRC522 Library

Used in Arduino to interface with the RFID module. It reads RFID card data and extracts the UID.

8.8.7 Keypad Library

Used to read input from the keypad. It detects key presses for PIN entry.

8.8.8 LiquidCrystal_I2C

Used to control the LCD display. It shows messages like "Enter PIN", "Access Granted", etc.

8.8.9 SoftwareSerial

Used for communication with the GSM module. It helps in sending SMS alerts during unauthorized access.

CHAPTER 9
METHODOLOGY

9.1 Dataset Creation with UID and PIN

Initially, a dataset of authorized users is created. For each user, multiple face images are collected and stored in separate folders labeled with their names. Along with the facial data, each user is assigned a unique RFID card and a corresponding PIN. The RFID card provides a unique UID for identification, and the PIN acts as an additional security layer. This combination of face, UID, and PIN forms a multi-level authentication dataset.

9.2 Face Encoding

The collected images are processed using Python code. The system detects faces using HOG or CNN methods and converts them into numerical encodings (128-dimensional vectors). Data augmentation techniques such as image flipping and brightness adjustment are applied to improve accuracy. These encodings are stored in a file named encodings.pkl, which is used for future face recognition.

9.3 Hardware Setup and Connections

All hardware components such as Arduino Uno, RFID module, keypad, LCD display, buzzer, and GSM module are connected using wires and a breadboard. The RFID module is connected using SPI communication, the keypad through digital pins, and the LCD using I2C interface. Proper connections ensure smooth communication between all components.

9.4 Arduino Code Development and Uploading

The Arduino code is written in Embedded C using Arduino IDE. Required libraries such as MFRC522, Keypad, LiquidCrystal_I2C, and SoftwareSerial are included to control the components. The code is compiled and uploaded to the Arduino board, enabling it to perform RFID reading, PIN verification, and communication tasks.

9.5 Serial Communication Setup

The Arduino is connected to the computer using a USB cable. Serial communication is established between Arduino and Python using PySerial. This allows both systems to exchange commands such as “START_FACE” and results like “GRANT” or “DENY”.

9.6 RFID Card Scanning

During operation, the user scans the RFID card using the RFID module. The system reads the unique UID

of the card.

9.7 RFID Verification

The scanned UID is compared with stored authorized UIDs. If the UID matches, the system proceeds to the next step. If not, access is denied and the system resets.

9.8 PIN Entry through Keypad

After RFID verification, the user is prompted to enter a PIN using the keypad. The entered PIN is displayed as '*' symbols on the LCD for security.

9.9 PIN Verification

The entered PIN is compared with the stored PIN. If correct, the system proceeds to face recognition; otherwise, access is denied.

9.10 Triggering Machine Learning Process

After successful RFID and PIN verification, the Arduino sends a command "START_FACE" to the Python program through serial communication to initiate face recognition.

9.11 Face Capture using Camera

The Python program uses OpenCV to activate the webcam and capture real-time video frames of the user.

9.12 Face Detection

The system detects the face from the captured frames using HOG or CNN methods, identifying the face region for further processing.

9.13 Face Recognition and Matching

The detected face is converted into an encoding and compared with stored encodings from encodings.pkl. If the similarity is within a threshold, the user is recognized.

9.14 Decision Making by ML System

Based on the comparison:

Match → "GRANT"

No match → "DENY"

9.15 Final Output and Alert System

The Arduino receives the result. If access is granted, it displays a success message on the LCD. If denied, the buzzer is activated and an SMS alert is sent using the GSM module.

9.16 System Reset

Finally, the system resets and waits for the next user, ensuring continuous operation.

Key Outcomes :

- Successful RFID UID detection
- Accurate PIN verification
- Real-time face recognition
- Proper GRANT/DENY decision
- SMS alert for unauthorized access

CHAPTER 10

WORKING PRINCIPLE

The proposed ATM fraud detection system operates based on a multi-level authentication mechanism that integrates embedded systems with machine learning techniques. The system ensures secure access by verifying the user through three stages: RFID authentication, PIN verification, and face recognition. This layered approach significantly reduces the chances of unauthorized access and enhances overall system security.

The working of the system begins when the user initiates the process by scanning the RFID card. Each RFID card is associated with a unique identification number (UID), which acts as the primary identifier for the user. When the card is brought near the RFID reader, the system reads the UID and sends it to the Arduino microcontroller. The Arduino then compares the scanned UID with the list of authorized UIDs stored in the system. If the UID matches, the system proceeds to the next stage; otherwise, access is immediately denied, preventing unauthorized entry at the initial level itself.

Once the RFID verification is successful, the system prompts the user to enter a Personal Identification Number (PIN) using the keypad. The keypad allows the user to input the PIN securely, and the entered digits are displayed as masked characters on the LCD screen to maintain confidentiality. The Arduino checks the entered PIN against the stored PIN associated with the scanned RFID card. If the PIN is correct, the system advances to the next stage of authentication. If the PIN is incorrect, access is denied, and the system resets, requiring the user to restart the process.

After successful RFID and PIN verification, the system activates the face recognition module, which is implemented using Python and machine learning libraries. At this stage, the Arduino sends a command "START_FACE" to the Python program via serial communication. This communication is established through a USB connection using the PySerial library, allowing seamless data exchange between the embedded system and the computer.

Upon receiving the command, the Python program activates the webcam using OpenCV and begins capturing real-time video frames of the user. The captured frames are processed to detect the presence of a face. The system uses HOG (Histogram of Oriented Gradients) or CNN (Convolutional Neural Network) methods for face detection. These techniques help identify the location of the face in the image by analyzing patterns such as edges and gradients.

Once the face is detected, the system extracts the facial features and converts them into a numerical representation known as a face encoding. This encoding is a 128-dimensional vector that uniquely represents the facial characteristics of the user. The generated encoding is then compared with the stored encodings of authorized users, which are saved in a file named encodings.pkl.

The comparison process is performed using a distance-based similarity measure. The system calculates the distance between the captured face encoding and the stored encodings. If the distance is below a predefined threshold value, the system considers it a match and identifies the user as authorized. If the distance exceeds the threshold, the system determines that the face does not match and treats it as an unauthorized attempt.

Based on the result of the face recognition process, the Python program sends a response back to the Arduino through serial communication. The response is either "GRANT" or "DENY". The Arduino reads this response and takes appropriate action. If the result is "GRANT", the system displays "Access Granted" on the LCD screen, indicating successful authentication. If the result is "DENY", the system activates the buzzer to alert nearby individuals and sends an SMS notification to a registered mobile number using the GSM module.

The GSM module communicates with the Arduino using SoftwareSerial and is programmed to send alert messages in case of unauthorized access attempts. This ensures that the account holder or concerned authority is immediately notified about any suspicious activity.

After completing the authentication process, the system resets itself to the initial state. All temporary data is cleared, and the system returns to the standby mode, waiting for the next user to scan their RFID card. This continuous cycle ensures smooth and efficient operation of the system.

Overall, the working principle of the proposed system combines hardware-based authentication with intelligent machine learning techniques to provide a secure, reliable, and efficient ATM fraud detection mechanism. By integrating RFID, PIN, and face recognition, the system offers a robust multi-layered security solution that minimizes the risk of fraud and unauthorized access.

Key Features

- RFID-based authentication
- PIN verification system
- Face recognition using ML
- Buzzer and GSM alert
- Real-time decision making

CHAPTER 11

FLOWCHART/ALGORITHM

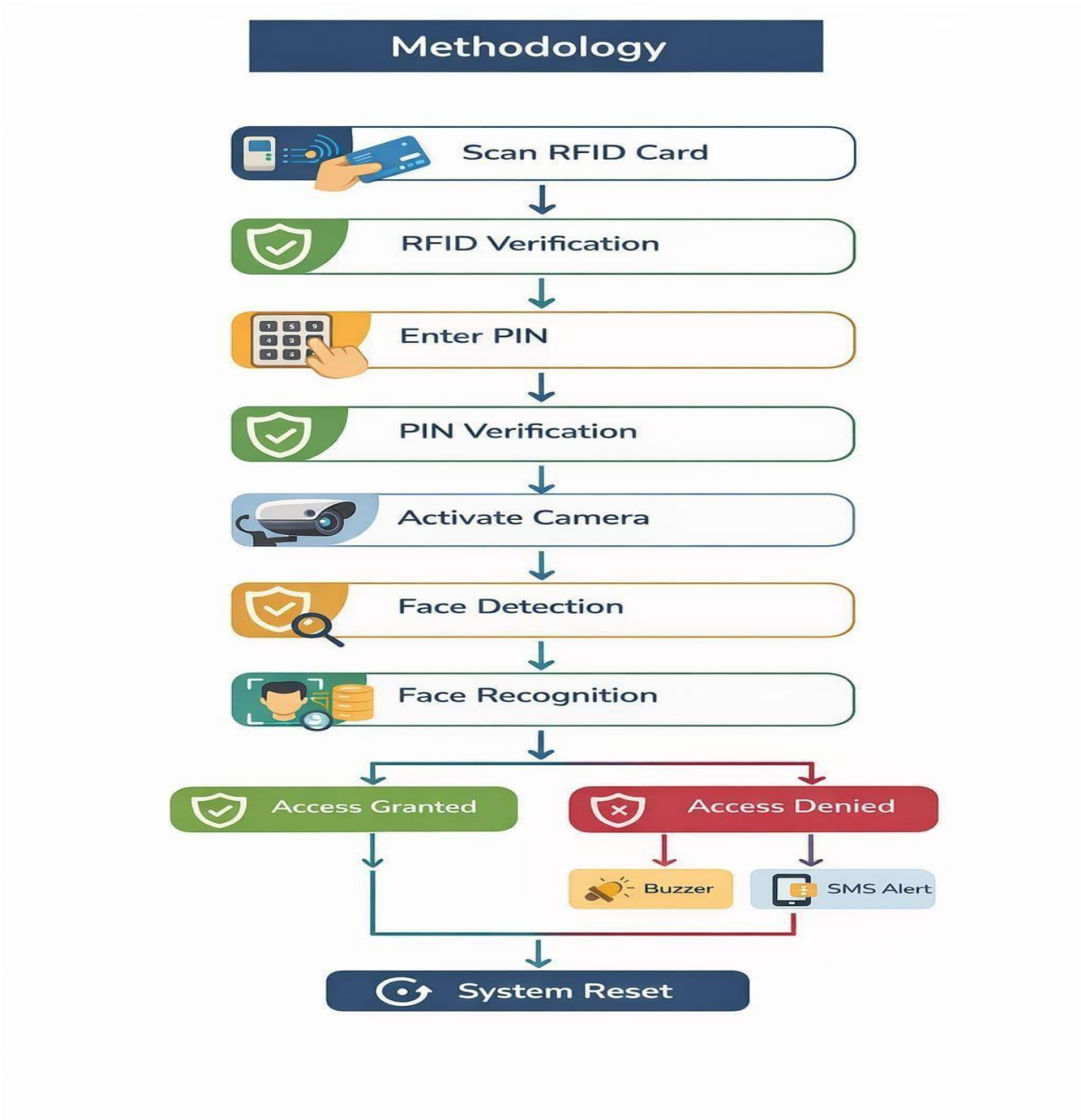

Figure 15 ***Fig11.1 Methodology Diagram***

CHAPTER 12

IMPLEMENTATION

The implementation of the proposed ATM fraud detection system involves the integration of embedded hardware components with machine learning-based software modules. The system is developed in two major parts: the hardware implementation using Arduino and the software implementation using Python for face recognition. Both parts are interconnected through serial communication to ensure smooth operation and real-time data exchange.

The implementation begins with the development of the dataset required for face recognition. Multiple facial images of authorized users are collected and stored in a structured format, where each user's images are placed in a separate folder labeled with their respective names. Along with facial data, each user is assigned a unique RFID card and a secure PIN, which are stored in the system. The facial images are processed using Python libraries such as OpenCV and face_recognition. During this process, faces are detected using HOG or CNN models and converted into numerical encodings. These encodings are 128-dimensional feature vectors that uniquely represent each face. To improve system accuracy, data augmentation techniques such as horizontal flipping and brightness adjustment are applied. The generated encodings are stored in a file named encodings.pkl, which acts as the database for user identification during authentication.

The hardware implementation is carried out using an Arduino Uno microcontroller, which acts as the central control unit. Various components such as the RFID module, keypad, LCD display, buzzer, and GSM module are connected to the Arduino. The RFID module is interfaced using SPI communication, while the keypad is connected to digital input pins for PIN entry. The LCD display is connected using the I2C interface to display messages such as "Scan RFID", "Enter PIN", and authentication results. The buzzer is used as an alert device, and the GSM module is connected using SoftwareSerial to send SMS notifications in case of unauthorized access.

The Arduino code is written in Embedded C using the Arduino IDE. Necessary libraries such as MFRC522, Keypad, LiquidCrystal_I2C, and SoftwareSerial are included to control the respective

hardware components. The code is designed to handle multiple operations including reading RFID card data, verifying the PIN, displaying messages on the LCD, and communicating with the Python-based face recognition system. Once the code is written and verified, it is compiled and uploaded to the Arduino board.

To enable communication between the Arduino and the Python program, serial communication is established using a USB cable. The Arduino uses serial functions such as Serial.begin() and Serial.println() to send data, while the Python program uses the PySerial library to receive and transmit data. This communication plays a crucial role in coordinating the embedded system with the machine learning module.

During the execution phase, the system starts by displaying a message on the LCD prompting the user to scan the RFID card. When the user scans the card, the RFID module reads the unique UID and sends it to the Arduino. The Arduino compares the scanned UID with the stored authorized UIDs. If the UID is valid, the system allows the user to proceed to the next step; otherwise, access is denied, and the system resets. After successful RFID verification, the system prompts the user to enter a PIN using the keypad. The entered PIN is masked on the LCD display for security purposes. The Arduino compares the entered PIN with the stored PIN associated with the RFID card. If the PIN is correct, the system proceeds to face recognition; otherwise, access is denied, and the system resets.

Once both RFID and PIN verifications are successful, the Arduino sends a command "START_FACE" to the Python program through serial communication. Upon receiving this command, the Python program activates the webcam using OpenCV and begins capturing real-time video frames. The system detects the face from the captured frames using HOG or CNN-based methods. After detecting the face, it is converted into a numerical encoding and compared with the stored encodings in the encodings.pkl file.

The comparison is performed using a distance-based method, where the similarity between the captured encoding and stored encodings is calculated. If the distance is below a predefined threshold, the system identifies the face as a match and sends a "GRANT" signal to the Arduino. If the distance exceeds the threshold, the system sends a "DENY" signal, indicating an unauthorized access attempt.

Upon receiving the result from the Python program, the Arduino performs the final action. If the result is "GRANT", the system displays "Access Granted" on the LCD and allows the user to proceed. If the result is "DENY", the system activates the buzzer to alert nearby individuals and sends an SMS notification using the GSM module. The SMS alert informs the registered user about the unauthorized access attempt, enhancing system security.

After completing the authentication process, the system resets itself to the initial state. All temporary data

is cleared, and the system returns to standby mode, waiting for the next user to scan their RFID card. This ensures continuous and efficient operation of the system.

Overall, the implementation of the ATM fraud detection system successfully integrates hardware and software components to provide a secure, reliable, and efficient authentication mechanism.

CHAPTER 13

RESULTS AND DISCUSSION

1. RFID Scanning Result

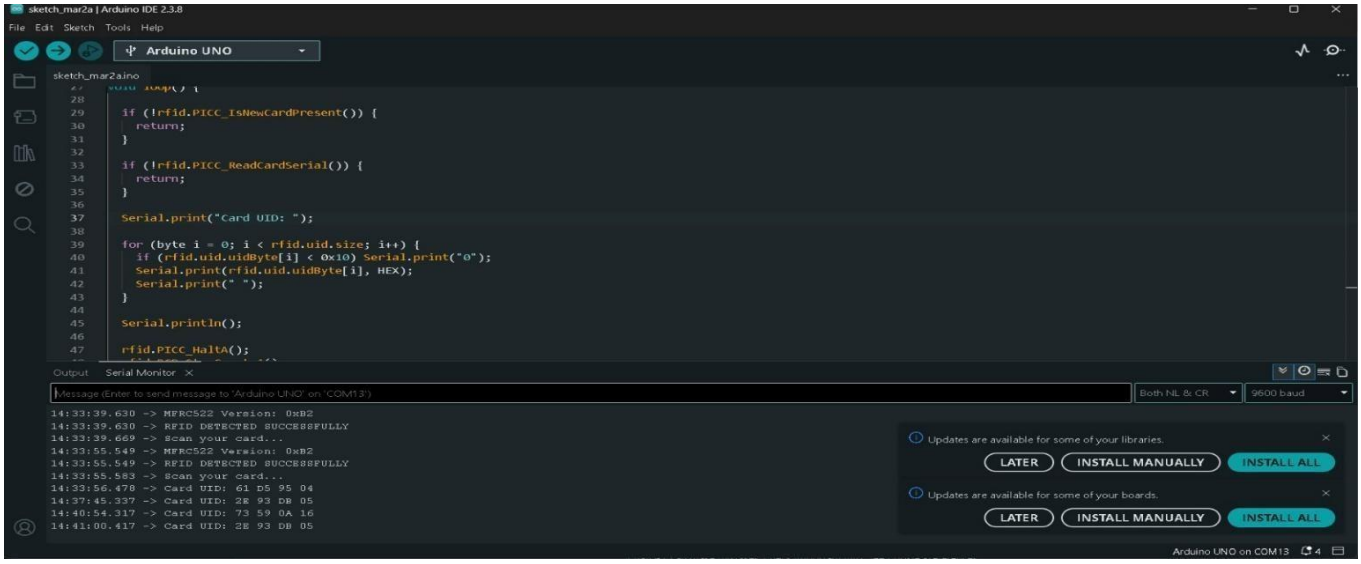

Figure 16

Fig 13.1 RFID UID

Explanation:

The RFID module successfully reads the unique UID of the scanned card when it is placed near the reader. The UID is displayed on the serial monitor or LCD screen and is compared with the stored authorized IDs. If the UID matches with the registered user, the system allows further authentication. If the UID is not recognized, access is denied immediately. This confirms that the RFID module works accurately as the first level of security in the system.

2. Pin Verification Result:

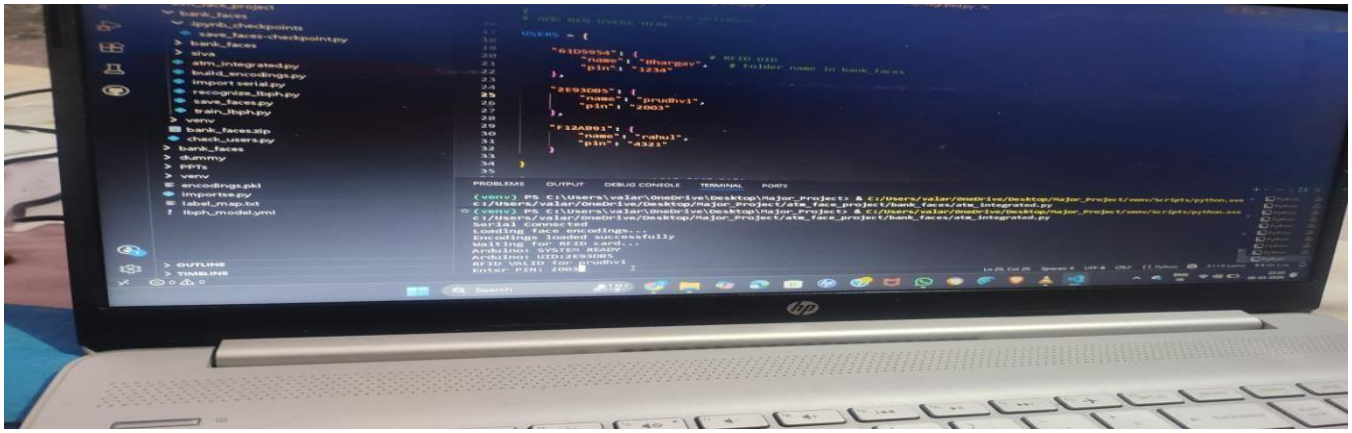

Figure 17

Fig 13.2 Pin Verification

Explanation:

After successful RFID verification, the system prompts the user to enter a PIN using the keypad. The entered PIN is displayed as masked characters on the LCD for security purposes. The system compares the entered PIN with the stored PIN. If the PIN is correct, the system proceeds to face recognition. Otherwise, access is denied. This step ensures an additional layer of security.

3. Face Detection and Recognition Result

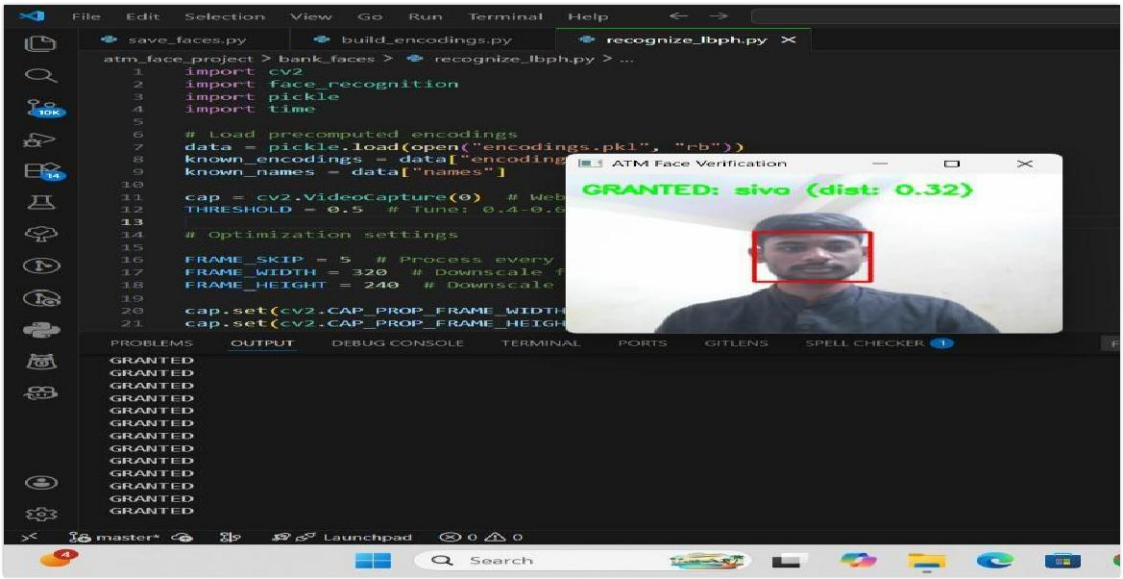

Figure 18 ***Fig 13.3 Face Detection and Recognition***

Explanation:

The system captures real-time video using a webcam and detects the face using HOG or CNN-based methods. The detected face is converted into an encoding and compared with stored encodings. If the face matches with the authorized user, the system displays “GRANTED”. If not, it displays “DENIED”. The results show that the system is capable of accurately recognizing authorized users under different conditions.

CHAPTER 14

Advantages of System

14.1 Multi-Level Security

One of the major advantages of the proposed system is its multi-level security mechanism. Unlike traditional ATM systems that rely only on a card and PIN, this system integrates three levels of authentication: RFID card verification, PIN validation, and face recognition. This layered approach significantly enhances security because even if one level is compromised, the remaining layers continue to protect the system. For example, if an unauthorized person obtains the RFID card, they still need to know the correct PIN and pass the face recognition stage. Similarly, even if someone guesses the PIN, they cannot bypass facial verification. This reduces the chances of fraud and unauthorized access to a great extent. The integration of machine learning further strengthens the system by ensuring that only the registered user's face is recognized. Overall, this multi-level authentication system provides a robust and reliable solution for ATM security, making it much more secure than conventional systems and ensuring that only genuine users can access the account.

14.2 Real-Time Fraud Detection and Alert System

Another important advantage of the system is its ability to detect fraud attempts in real time and immediately notify the concerned user. Whenever an unauthorized access attempt is detected—such as an incorrect PIN entry or face mismatch—the system instantly activates a buzzer to alert nearby individuals. At the same time, it sends an SMS notification to the registered mobile number using the GSM module. This immediate response helps in preventing further misuse and allows the user to take quick action. Real-time alerts are crucial in financial systems like ATMs, where even a small delay can lead to significant losses. By providing instant notifications, the system ensures that the account holder is always aware of suspicious activities. This feature not only enhances security but also increases user confidence in the system. The combination of local alerts (buzzer) and remote alerts (SMS) makes the system highly effective in detecting and responding to fraud attempts promptly.

14.3 High Accuracy using Machine Learning

The use of machine learning techniques, particularly face recognition, provides high accuracy in identifying authorized users. The system uses advanced algorithms such as HOG and CNN to detect and recognize faces. Each face is converted into a unique encoding, which is then compared with stored encodings to determine a match. This process ensures that even slight variations in appearance, such as changes in lighting, angle, or facial expressions, can be handled effectively. Data augmentation techniques further improve the accuracy by training the system with varied images. As a result, the system can reliably

distinguish between authorized and unauthorized users. High accuracy is essential in security systems to avoid both false acceptance and false rejection. By leveraging machine learning, the proposed system achieves a balance between security and usability. This makes it a powerful solution for real-world applications where precise identification is critical.

14.4 User-Friendly and Easy to Operate

Despite incorporating advanced technologies, the system is designed to be simple and user-friendly. The process of authentication—scanning an RFID card, entering a PIN, and facing the camera—is easy to understand and similar to existing ATM operations. The LCD display provides clear instructions at each step, guiding the user throughout the process. The keypad allows easy PIN entry, while the system automatically handles complex operations like face recognition in the background. This ensures that users do not need any technical knowledge to operate the system. Additionally, the system provides immediate feedback, such as "Access Granted" or "Access Denied," which helps users understand the result quickly. A user-friendly design is important for widespread adoption, especially in systems like ATMs that are used by people of all age groups. Therefore, the proposed system successfully combines advanced security features with ease of use.

14.5 Cost-Effective and Scalable Solution

The proposed system is built using affordable and easily available components such as Arduino, RFID module, keypad, and camera. This makes it a cost-effective solution compared to high-end biometric systems. Despite its low cost, the system provides strong security by integrating hardware and software components effectively. Additionally, the system is scalable, meaning it can be easily expanded to support more users by adding their data to the dataset. The machine learning model can also be updated with new images to improve performance over time. This flexibility makes the system suitable for various applications beyond ATMs, such as secure access control in offices, banks, and restricted areas. The combination of low cost and scalability makes the system practical for real-world implementation, especially in environments where budget constraints are a concern. Overall, the system offers an efficient and economical solution without compromising on security.

CHAPTER 15

LIMITATIONS

15.1 Dependence on Lighting and Environmental Conditions

One of the major limitations of the proposed system is its dependency on environmental conditions, especially lighting. The face recognition module relies on capturing clear facial features using a camera. If the lighting conditions are too low, too bright, or uneven, the system may fail to detect the face accurately. Shadows, reflections, and backlighting can also affect the quality of the captured image, leading to incorrect face detection or recognition errors. Additionally, changes in a user's appearance, such as wearing masks, glasses, or changes in hairstyle, can impact recognition accuracy. Although techniques like data augmentation and advanced algorithms such as CNN improve performance, they cannot completely eliminate these challenges. In real-world ATM environments, where lighting conditions may vary throughout the day or across locations, this limitation can affect system reliability. Therefore, proper camera placement and controlled lighting conditions are required to achieve optimal performance, which may not always be feasible in practical scenarios.

15.2 Processing Time and System Delay

Another limitation of the system is the processing time required for face recognition. The machine learning module, especially when using CNN-based methods, requires significant computational resources to when running on systems with limited hardware capabilities. The delay may affect user experience, as ATM users generally expect quick transactions. Additionally, real-time video processing involves capturing multiple frames, detecting faces, and performing comparisons, which further increases processing time. If the system takes too long to respond, users may find it inconvenient and inefficient. While optimization techniques and faster hardware can reduce delays, they may increase the overall cost of the system. Therefore, there is a trade-off between accuracy and speed. Ensuring a balance between quick response time and reliable face recognition remains a challenge in the implementation of such systems.

15.3 Dependency on Hardware and Connectivity

The system heavily depends on multiple hardware components such as RFID module, keypad, camera, GSM module, and Arduino. If any one of these components fails or malfunctions, the entire system may not function properly. For example, if the RFID module fails, the user cannot initiate the authentication process. Similarly, if the camera is not working, face recognition cannot be performed. The GSM module also requires network connectivity to send SMS alerts, and in areas with poor signal strength, the alert

system may fail. Additionally, the system depends on continuous communication between Arduino and the Python program through serial communication. Any interruption in this connection can lead to incomplete or failed authentication. This dependency on multiple components increases the complexity of the system and may require regular maintenance. Hence, hardware reliability and stable connectivity are critical for the proper functioning of the system, which can be a limitation in real-world deployment.

15.4 Limited Scalability and Data Management

Although the system is scalable to some extent, managing a large number of users can become challenging. As more users are added, the size of the dataset and the number of stored encodings increase. This can slow down the face recognition process, as the system needs to compare the captured encoding with a larger database. Additionally, storing and managing large datasets requires proper organization and memory management. The current implementation uses a file-based approach (encodings.pkl), which may not be efficient for handling a very large number of users. In large-scale applications such as banking networks with thousands of users, more advanced database systems and optimized algorithms would be required. Without such improvements, the system may face performance issues and delays. Therefore, while the system works efficiently for small to medium-scale applications, its scalability is limited in its current form.

15.5 Security and Privacy Concerns

The use of biometric data such as facial images introduces concerns related to security and privacy. Facial data is sensitive information, and if it is not stored securely, it can be misused. In the current system, face encodings are stored in a local file, which may be vulnerable to unauthorized access if proper security measures are not implemented. Additionally, users may have concerns about their facial data being collected and stored, especially if the system is not transparent about data usage. There is also a risk of spoofing attacks, where an attacker may attempt to use a photograph or video of an authorized user to bypass the system. Although advanced techniques can reduce such risks, they cannot be completely eliminated. Therefore, ensuring data security, encryption, and user privacy is essential. Addressing these concerns is important for building trust and ensuring the safe deployment of the system in real-world applications.

CHAPTER 16

APPLICATIONS

16.1 Preventing Unauthorized Access

ATM fraud detection systems play a crucial role in preventing unauthorized access to bank accounts. These systems use technologies such as RFID card scanning, PIN verification, and biometric authentication (fingerprint or face recognition) to ensure that only the legitimate account holder can perform transactions. By identifying suspicious attempts—such as multiple failed PIN entries or the use of cloned cards—the system can immediately block the transaction and alert the bank. This proactive approach reduces the likelihood of financial loss for both customers and banks. Moreover, ATM fraud detection systems continuously log transaction patterns, which can help in identifying abnormal access behaviors. For example, if an ATM card is suddenly used in a different city or country, the system can flag it as suspicious. These measures not only safeguard individual accounts but also strengthen the overall security of the banking network, maintaining trust between banks and customers. Implementing such preventive measures is especially important in high-risk areas where ATM skimming and card cloning are common. With real-time monitoring, unauthorized attempts can be stopped before any financial damage occurs, making ATM fraud detection essential for modern banking security.

16.2 Detection of Skimming and Cloning Attacks

One of the most common types of ATM fraud is card skimming, where criminals attach devices to ATMs to capture card information. ATM fraud detection systems can identify these attacks by monitoring the behavior of cards and machines. For example, sensors can detect tampered card slots or unusual data patterns during transactions. Advanced systems integrate machine learning algorithms to analyze transaction trends, enabling them to recognize anomalies like repeated card declines, unusual withdrawal amounts, or multiple cards being used at the same terminal. Once such suspicious activity is detected, the system can immediately notify bank authorities or disable the ATM temporarily. This not only prevents the loss of customer funds but also allows banks to investigate and secure compromised ATMs quickly. Additionally, integrating biometric verification ensures that even if card data is cloned, unauthorized users cannot access the account. The ability to detect and respond to skimming and cloning attacks in real time significantly reduces the potential financial and reputational damage to banks and protects customers from becoming victims of fraud.

16.3 Enhancing Customer Trust and Satisfaction

ATM fraud detection systems are not just about security—they also enhance customer trust in banking services. Customers today expect their financial transactions to be secure, convenient, and reliable. By

implementing robust fraud detection measures, banks reassure users that their money is protected against unauthorized access and malicious activities. Features like real-time alerts, biometric authentication, and transaction monitoring give customers confidence that any unusual activity will be detected and addressed promptly. This trust encourages users to engage more actively with banking services, including online and mobile banking, which can improve the bank's overall business performance. Moreover, in case of a detected fraud attempt, the customer is immediately informed, allowing them to take preventive action, such as changing their PIN or temporarily blocking the card. Maintaining high security standards also reduces the number of customer complaints and disputes, saving banks time and resources in resolving issues. Ultimately, ATM fraud detection systems contribute to a safer, smoother, and more reliable banking experience, which strengthens customer loyalty and the institution's reputation.

16.4 Fraud Analytics and Reporting

ATM fraud detection systems serve as powerful tools for fraud analytics and reporting. By continuously collecting transaction data from multiple ATMs, these systems provide banks with insights into patterns of suspicious activity. For instance, analytics can reveal high-risk locations, frequent attempts of card cloning, or unusual withdrawal patterns that deviate from a customer's typical behavior. This information allows banks to make data-driven decisions to strengthen security measures, such as upgrading ATM hardware, improving surveillance, or revising transaction limits. Reporting tools also enable banks to comply with regulatory requirements, providing authorities with detailed evidence of attempted fraud. Over time, this data helps in predicting future fraudulent trends, making prevention more proactive rather than reactive. Additionally, machine learning algorithms can analyze historical fraud cases to identify subtle signs of potential threats, increasing the system's efficiency in detecting fraud. By leveraging analytics and reporting, banks can not only minimize financial losses but also optimize operational processes, focusing on ATMs or areas that are most vulnerable. This enhances the overall effectiveness of fraud management while providing actionable intelligence for security planning and decision-making.

16.5 Regulatory Compliance and Risk Management

ATM fraud detection is essential for regulatory compliance and risk management in the banking sector. Financial institutions are required by law to implement security measures that protect customer funds and report fraudulent activities. By using advanced ATM fraud detection systems, banks can meet these regulatory requirements efficiently. These systems monitor transactions in real time, record suspicious activities, and generate automated reports that comply with legal standards. Additionally, they help in risk management by identifying vulnerabilities in ATM networks, such as weak points susceptible to card skimming, malware attacks, or insider threats. By mitigating these risks proactively, banks reduce

potential financial liabilities and maintain operational integrity. Regulatory authorities often audit banks to ensure that fraud detection measures are in place, making these systems a critical component of compliance. Moreover, demonstrating robust fraud prevention practices strengthens the bank's credibility with regulators, investors, and customers.

CHAPTER 17

Future Scope

The future of ATM fraud detection is increasingly moving towards intelligent, automated systems powered by deep learning and artificial intelligence. As fraudsters adopt more sophisticated techniques such as card skimming, PIN theft, and synthetic identity attacks, traditional security measures like static PIN verification or CCTV monitoring are no longer sufficient. Deep learning-based ATM fraud detection systems have the potential to revolutionize banking security by automatically learning patterns of normal customer behavior and detecting anomalies in real time. For example, neural networks can analyze transaction sequences, withdrawal amounts, frequency, and location data to identify suspicious behavior, even before an attack is completed.

One of the most promising directions is the integration of multimodal biometric authentication. Deep learning models can be trained to process facial recognition, fingerprint scanning, and iris detection simultaneously, providing a highly accurate and secure method for verifying a customer's identity. Coupled with real-time video analytics, the system can detect physical tampering or unusual user behavior at the ATM, adding an extra layer of security.

Moreover, the system can evolve into a predictive fraud prevention platform. By continuously learning from historical data and adapting to new attack patterns, deep learning algorithms can anticipate potential fraudulent activities and trigger automated preventive actions. This includes temporarily blocking transactions, alerting bank authorities, or notifying the account holder through mobile and online banking channels. The incorporation of IoT-enabled smart ATMs will further enhance this capability, as sensors can monitor tampering attempts, card skimmers, or unusual device interactions, feeding this data into deep learning models for instant analysis.

In addition, deep learning-based ATM fraud detection can support centralized monitoring and risk management. Banks can implement cloud-connected systems that aggregate transaction data from multiple ATMs, allowing AI models to detect broader fraud trends across regions or networks. This not only improves response times but also helps banks comply with regulatory standards and provides detailed forensic evidence in case of criminal investigations.

Looking ahead, the future scope includes fully autonomous, adaptive, and self-learning ATM security platforms. These systems will continuously improve with exposure to new fraud scenarios, ensuring that even unknown attack methods are identified and mitigated effectively. By combining deep learning with IoT, biometrics, and predictive analytics, ATM fraud detection can achieve near real-time security, minimize financial losses, enhance customer trust, and set a new standard for the global banking .

CHAPTER 18

CODES

18.1 Code For Saving Faces

```
import cv2
import os

# Change 'user2' to the new user name
name = "prudhvi"
out_dir = f"bank_faces/{name}"
os.makedirs(out_dir, exist_ok=True)

cap = cv2.VideoCapture(0) # Webcam
i = 0
print("Press 's' to save each photo. Capture 100 images. Press 'q' to quit early.")

while i < 10:  # Increased for high accuracy
    ret, frame = cap.read()
    if not ret:
        print("Error: Webcam not accessible.")
        break
    cv2.imshow("Capture (press 's' to save)", frame)
    key = cv2.waitKey(1) & 0xFF
    if key == ord('s'):
        cv2.imwrite(f"{out_dir}/{name}_{i}.jpg", frame)
        print(f"Saved {name}_{i}.jpg")
        i += 1
    elif key == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
print(f"Done! Captured {i} images for {name}.")
```

18.2 Code To Build Encoding

```
import face_recognition
import cv2
import os
import pickle
import numpy as np

DATA_DIR = "bank_faces"
AUGMENT_FACTOR = 2  # Augment each image (flip + brightness)

# Manually set the target user here (change this value as needed)
```

```
TARGET_USER = "prudhvi"  # Example: Change to "user2" or "user3"

# Load existing encodings if file exists
if os.path.exists("encodings.pkl"):
    with open("encodings.pkl", "rb") as f:
        data = pickle.load(f)
    known_encodings = data["encodings"]
    known_names = data["names"]
    print(f"Loaded existing {len(known_encodings)} encodings for {len(set(known_names))} users.")
else:
    known_encodings = []
    known_names = []

# Check if target user already exists
if TARGET_USER in known_names:
    print(f"User '{TARGET_USER}' already exists. Skipping training.")
else:
    person_dir = os.path.join(DATA_DIR, TARGET_USER)
    if not os.path.isdir(person_dir):
        print(f"Folder '{person_dir}' not found. Create it and add images.")
    else:
        print(f"Processing new user: {TARGET_USER}")
        user_encodings = []
        for filename in os.listdir(person_dir):
            if filename.endswith('.jpg', '.png')):
                filepath = os.path.join(person_dir, filename)
                img = cv2.imread(filepath)
                if img is None:
                    print(f"Warning: Could not load {filepath}")
                    continue
                rgb_img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

                # Original image
                boxes = face_recognition.face_locations(rgb_img, model="cnn")  # High accuracy; use "hog"
if slow
                if boxes:
                    encodings = face_recognition.face_encodings(rgb_img, boxes)
                    if encodings:
                        user_encodings.append(encodings[0])

                # Augmentation 1: Horizontal flip
                flipped = cv2.flip(rgb_img, 1)
```

```
            boxes_flip = face_recognition.face_locations(flipped, model="cnn")
            if boxes_flip:
                encodings_flip = face_recognition.face_encodings(flipped, boxes_flip)
                if encodings_flip:
                    user_encodings.append(encodings_flip[0])

            # Augmentation 2: Brightness adjustment (simulate lighting variation)
            bright_img = cv2.convertScaleAbs(rgb_img, alpha=1.2, beta=10)  # Increase brightness
            boxes_bright = face_recognition.face_locations(bright_img, model="hog")
            if boxes_bright:
                encodings_bright = face_recognition.face_encodings(bright_img, boxes_bright)
                if encodings_bright:
                    user_encodings.append(encodings_bright[0])

            print(f"Processed {filepath} with {AUGMENT_FACTOR} augmentations")

        # Add new user's encodings (multiple per image due to augmentation)
        for enc in user_encodings:
            known_encodings.append(enc)
            known_names.append(TARGET_USER)

# Save updated encodings
with open("encodings.pkl", "wb") as f:
    pickle.dump({"encodings": known_encodings, "names": known_names}, f)

print(f"Saved updated encodings.pkl with {len(known_encodings)} total encodings for
{len(set(known_names))} users.")
```

18.3 Integration Code

```
import cv2
import face_recognition
import pickle
import serial
import time

# ---------------- SERIAL CONNECTION ----------------

ser = serial.Serial("COM13",9600,timeout=1)
time.sleep(2)

print("Serial Connected")

# ------------------ USER DATABASE ------------------
```

```
USERS = {

    "61D5954":{
        "name":"Bhargav",
        "pin":"1234"
    },

    "2E93DB5":{
        "name":"sampath",
        "pin":"2003"
    },

    "F12AB91":{
        "name":"rahul",
        "pin":"4321"
    }

}

# ------------------ LOAD FACE DATA ------------------

print("Loading face encodings...")

data = pickle.load(open("encodings.pkl","rb"))

known_encodings = data["encodings"]
known_names = data["names"]

print("Encodings loaded successfully")

# ---------------- FACE VERIFICATION ----------------

def face_verification(target_name):

    cap = cv2.VideoCapture(0)

    THRESHOLD = 0.5
    FRAME_SKIP = 5

    frame_count = 0
    grant_count = 0
    verdict = "DENIED"

    print("Starting Face Verification for",target_name)

    while True:

        ret,frame = cap.read()
```

```
        if not ret:
            break

        frame_count += 1

        if frame_count % FRAME_SKIP == 0:

            rgb = cv2.cvtColor(frame,cv2.COLOR_BGR2RGB)

            face_locations = face_recognition.face_locations(rgb)
            face_encodings = face_recognition.face_encodings(rgb,face_locations)

            verdict = "DENIED"

            if face_encodings:

                distances = face_recognition.face_distance(
                    known_encodings,face_encodings[0])

                min_distance = min(distances)
                index = distances.argmin()

                if min_distance < THRESHOLD and known_names[index] == target_name:

                    verdict = "GRANTED"
                    grant_count += 1

        cv2.putText(frame,verdict,(20,40),
                    cv2.FONT_HERSHEY_SIMPLEX,1,
                    (0,255,0) if verdict=="GRANTED" else (0,0,255),2)

        cv2.imshow("ATM Face Verification",frame)

        if cv2.waitKey(1) & 0xFF == ord('q'):
            break

        if grant_count >= 5:

            cap.release()
            cv2.destroyAllWindows()
            return "GRANT"

    cap.release()
    cv2.destroyAllWindows()

    return "DENY"
# ------------------ ATM SYSTEM ------------------
```

```
print("Waiting for RFID + PIN...")

while True:

    if ser.in_waiting > 0:

        message = ser.readline().decode().strip()

        print("Arduino:",message)

        # Arduino sends UID first
        if message.startswith("UID:"):

            uid = message.replace("UID:","").upper()

            # wait for PIN message
            while True:

                if ser.in_waiting > 0:

                    pin_msg = ser.readline().decode().strip()

                    if pin_msg.startswith("PIN:"):

                        entered_pin = pin_msg.replace("PIN:","")
                        break

            print("UID:",uid)
            print("PIN:",entered_pin)

            if uid in USERS:

                user = USERS[uid]

                if entered_pin == user["pin"]:

                    print("PIN CORRECT")

                    result = face_verification(user["name"])

                    if result == "GRANT":

                        print("ACCESS GRANTED")
                        ser.write(b"LED_ON\n")

                    else:
```

```
            print("FACE NOT MATCHED")
            ser.write(b"FRAUD\n")

        else:

            print("WRONG PIN")
            ser.write(b"FRAUD\n")

    else:

        print("INVALID CARD")
        ser.write(b"FRAUD\n")
```

18.4 ARDUINO CODE

```
#include <SPI.h>
#include <MFRC522.h>
#include <Keypad.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

/* --------- RFID --------- */

#define SS_PIN 10
#define RST_PIN 9

MFRC522 rfid(SS_PIN, RST_PIN);

/* --------- LCD --------- */

LiquidCrystal_I2C lcd(0x27,16,2);

/* -------- KEYPAD --------- */

const byte ROWS = 4;
const byte COLS = 4;

char keys[ROWS][COLS] =
{
{'1','2','3','A'},
{'4','5','6','B'},
{'7','8','9','C'},
{'*','0','#','D'}
};

byte rowPins[ROWS] = {2,3,4,5};
byte colPins[COLS] = {A0,A1,A2,A3};

Keypad keypad = Keypad(makeKeymap(keys),rowPins,colPins,ROWS,COLS);
```

```
/* -------- OUTPUT ---------*/

#define LED 6
#define BUZZER 7

String enteredPIN = "";
String uid = "";

void setup()
{

Serial.begin(9600);

SPI.begin();
rfid.PCD_Init();

lcd.init();
lcd.backlight();

pinMode(LED,OUTPUT);
pinMode(BUZZER,OUTPUT);

lcd.setCursor(0,0);
lcd.print("ATM SYSTEM");

lcd.setCursor(0,1);
lcd.print("Scan Card");

delay(2000);
lcd.clear();

}

/* -------- MAIN LOOP--------- */

void loop()
{

/* -------- WAIT FOR RFID --------- */

if(!rfid.PICC_IsNewCardPresent())
return;

if(!rfid.PICC_ReadCardSerial())
return;

/* -------- GET UID ---------*/
```

```
uid="";

for(byte i=0;i<rfid.uid.size;i++)
{
uid += String(rfid.uid.uidByte[i],HEX);
}

uid.toUpperCase();

/* -------- DISPLAY -------- */

lcd.clear();
lcd.setCursor(0,0);
lcd.print("Card Detected");

/* -------- SEND UID ---------*/

Serial.print("UID:");
Serial.println(uid);

delay(1000);

/* -------- ENTER PIN --------- */

lcd.clear();
lcd.setCursor(0,0);
lcd.print("Enter PIN:");

enteredPIN="";

while(enteredPIN.length()<4)
{

char key = keypad.getKey();

if(key)
{

enteredPIN += key;

lcd.setCursor(enteredPIN.length()-1,1);
lcd.print("*");

}

}
```

```
/* -------- SEND PIN--------- */

Serial.print("PIN:");
Serial.println(enteredPIN);

/* -------- WAIT FOR PYTHON--------- */

lcd.clear();
lcd.print("Verifying. ");

while(!Serial.available());

String result = Serial.readStringUntil('\n');
result.trim();

/* -------- ACCESS GRANTED--------- */

if(result == "LED_ON")
{

lcd.clear();
lcd.print("ACCESS GRANTED");

digitalWrite(LED,HIGH);

delay(5000);

digitalWrite(LED,LOW);

}

/* ------ FRAUD------ */

else if(result == "FRAUD")
{

lcd.clear();
lcd.print("FRAUD ALERT");

for(int i=0;i<5;i++)
{
digitalWrite(BUZZER,HIGH);
delay(300);
digitalWrite(BUZZER,LOW);
delay(300);
}

}
```

```
delay(3000);

/* -------- RESET SYSTEM ---------*/

lcd.clear();
lcd.print("Scan Card");

rfid.PICC_HaltA();

}
```

CHAPTER 19

CONCLUSION

With the increasing reliance on ATMs for everyday banking, these machines have become a primary target for fraudsters using techniques such as card skimming, PIN theft, and identity fraud. Traditional security measures are no longer sufficient to protect customer assets. This project presents a comprehensive ATM fraud detection system that integrates facial recognition, RFID authentication, PIN verification, and GSM alerts, powered by deep learning algorithms. By combining multiple layers of authentication, the system ensures that only legitimate users can access accounts, while significantly reducing the risk of unauthorized transactions.

Facial recognition serves as the primary identity verification method, analyzing the user's facial features in real time to confirm their identity. RFID adds an additional verification layer by checking the unique card signal, while the PIN provides a knowledge-based authentication step. Together, these layers create a robust, multi-factor security framework that is extremely difficult for fraudsters to bypass. Deep learning algorithms continuously analyze transaction patterns, user behavior, and facial data to detect anomalies and identify suspicious activities before financial loss occurs.

A key feature of this system is the GSM-based alert mechanism, which immediately notifies the customer and bank authorities whenever a potential fraud is detected. This real-time communication allows for instant preventive action, such as temporarily blocking the card or stopping the transaction, minimizing risk and potential loss. Predictive analysis of ATM usage patterns further enhances security by helping banks anticipate fraud trends and take proactive measures to protect their network.

This integrated approach ensures both security and convenience. Customers can confidently use ATMs knowing their accounts are protected by multiple authentication layers and that they will be instantly informed of any suspicious activity. Banks benefit from reduced fraudulent activity, lower financial losses, and improved compliance with regulatory requirements. Moreover, the modular design of the system allows future upgrades, such as cloud-based monitoring, mobile banking integration, and IoT-enabled smart ATMs, creating a centralized, adaptive, and intelligent security network.

In conclusion, the ATM fraud detection system using facial recognition, RFID, PIN, and GSM alerts with deep learning provides a proactive, intelligent, and user-friendly solution to combat ATM fraud. By combining multi-factor authentication with real-time monitoring and alerting, the system protects customer assets, builds trust in banking services, and ensures that ATMs remain secure, reliable, and resilient against evolving threats.

CHAPTER 20

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Project Title: ATM FRAUD DETECTION SYSTEM

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Project Course Outcomes:

CO1: Understand and apply fundamental concepts related to smart shopping trolley systems, including RFID technology, automation, and sensor integration.

CO2: Analyze and implement hardware and software components, focusing on ESP 8266 for effective system development.

CO3: Evaluate and validate system performance, addressing key challenges in real-time implementation and customer usability.

CO4: Demonstrate the effectiveness of engineering principles in automation and suggest future enhancements for smart trolley performance.

CO-PO Mapping:

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	3	2	-	2	-	-	-	-	1	3	-	-	3
CO2	3	3	2	2	3	2	-	-	-	-	3	-	-	3
CO3	3	2	1	1	3	3	3	3	3	-	3	2	-	3
CO4	3	1	1	-	-	3	3	3	3	3	3	-	3	3

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